



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

PRE - SCORE

20-11-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS, TNAS, TAAS & TTAS)

ANSWER KEY

PAPER-1 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	D	A	B	C		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,C	D	B			
SECTION-I (iii)	Q.	8	9	10	11		
	A.	C	A	A	D		
SECTION-II	Q.	1	2	3	4	5	6
	A.	2	2	4	42	2	1

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	C	C	C	B		
SECTION-I (ii)	Q.	5	6	7			
	A.	B,C,D	A,B,D	B,D			
SECTION-I (iii)	Q.	8	9	10	11		
	A.	C	A	D	A		
SECTION-II	Q.	1	2	3	4	5	6
	A.	5	2	4	2	5	4

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	A	C	A		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,C,D	A,B,D	B,C,D			
SECTION-I (iii)	Q.	8	9	10	11		
	A.	C	C	B	A		
SECTION-II	Q.	1	2	3	4	5	6
	A.	1	8	1	6	2	324

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. **Ans (D)**

$$F = mg - F_z = mg - \frac{\pi R^2 P}{c}$$

and from the law of conservation of energy, we

can determine the sought velocity as

$$v = \sqrt{\frac{2FH}{m}} = \sqrt{2H \left(g - \frac{\pi R^2 P}{mc} \right)}$$

2. **Ans (A)**

If we denote the refractive index of the liquid

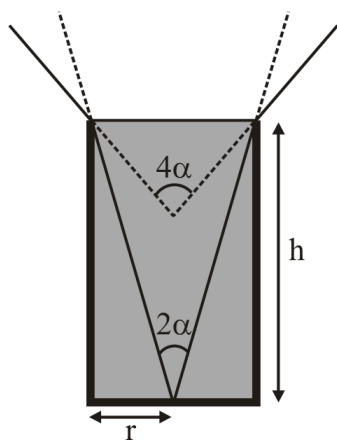
by n , Snell's law implies $\sin 2\alpha = n \sin \alpha$

Using the double angle formula $\sin 2\alpha = 2 \sin \alpha$

$\cos \alpha$ for the left hand side, $n = 2 \cos \alpha$

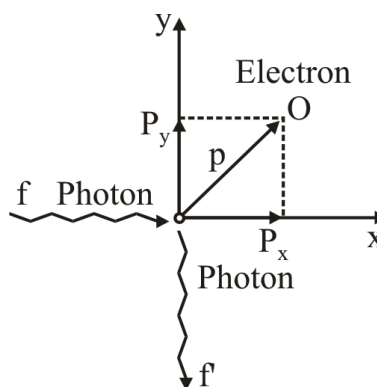
That's the boundary condition, so we need n

with greater or equal value, $n \geq 2 \cos \alpha$



4. **Ans (C)**

The energy of a photon of frequency f is hf , while its momentum is hf/c , where h is Planck's constant and c the velocity of light. The energy of the photon decreases to hf' after the collision, and its momentum, of magnitude hf'/c , is perpendicular to the original momentum.



It follows from momentum conservation that, as shown in the figure, the components of the electron's final momentum are $p_x = hf/c$ and $p_y = hf'/c$.

Using the principle of conservation of energy in its relativistic form, $E_2 = E_0^2 + p^2 c^2$, where E_0 is the rest-mass energy, we obtain

$$hf + E_0 = hf' + \sqrt{E_0^2 + (p_x^2 + p_y^2)c^2}.$$

The above relationships yield the change in the wavelength of the photon as

$$\Delta\lambda = \lambda' - \lambda = \frac{c}{f'} - \frac{c}{f} = \frac{hc}{E_0} = \frac{h}{mc}$$

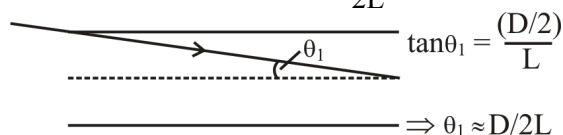
PART-1 : PHYSICS

SECTION-I (ii)

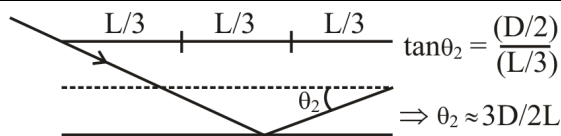
5. Ans (A,C)

When approaching this question, it may help to first sketch a diagram of the tube, with the centre marked as a dotted line, in preparation for drawing ray diagrams. Analyse first the light that is directly incident at the eye from the diffuse light source without reflection, then consider the first-order reflection, second-order reflection, and so on.

We start by drawing the ray that subtends the highest angle at the eye (which is at the centre of the tube). We see that where the diameter of the tube is $-D$, and where the length of the tube is L , the highest angle that can be subtended at the eye without reflection is given by $\tan\theta_1 = (D/2) / L$. For a long tube, taking the small-angle approximation, we have $\theta \approx \frac{D}{2L}$.

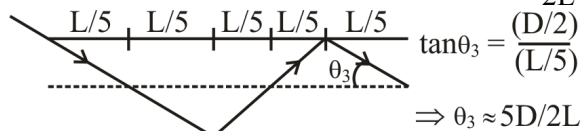


We then consider the first reflected ray. To do this, the only additional thing we need to know is the law of reflection, which is simply that the incident and reflected rays make the same angle with the wall. We see that the ray that subtends the highest angle at the eye, and which is reflected once, divides the tube into three. We have $\tan\theta_2 = (D/2) / (L/3)$. That is, $\theta_2 \approx \frac{3D}{2L}$.



We then consider the second reflected ray. The ray that subtends the highest angle at the eye, and which is reflected twice, divides the tube into five.

We have $\tan\theta_3 = (D/2) / (L/5)$. That is, $\theta_3 \approx \frac{5D}{2L}$.



We quickly see that the general formula for the angle subtended at the eye by the n -th ring is

$$\tan\theta_n = \frac{(D/2)}{(L/2n-1)}$$

Taking the small-angle approximation, we see that

$$\theta_n \approx \frac{(2n-1)D}{2L}$$

6. Ans (D)

$$L = m_e r v = n \hbar$$

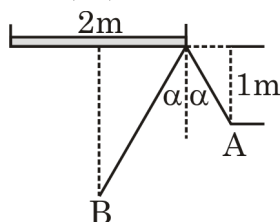
From this equation, we express r and insert it into the previous equation. This gives us

$$v = \frac{Ze^2}{4\pi\epsilon_0 n \hbar}$$

After substituting for $v = c$, $n = 1$ and expressing Z . We obtain

$$Z = \frac{4\pi\epsilon_0 \hbar c}{e^2}$$

7. Ans (B)



From similar triangles we get

$$\frac{1}{3.5-t} = \frac{2-t}{1}$$

We get a quadratic equation that has the solutions $t = 1.5$ s and $t = 4.0$ s where the first of them is the answer.

PART-1 : PHYSICS

SECTION-II

1. Ans (2)

First, let's look at how the beam passes through two lenses placed one behind the other. Let a be the distance of the object in front of the first lens with focal length f_1 . Just behind it is a second lens with focal length f_2 . Using the Gaussian lens equation, we find the position of the image a' of the object as

$$a' = \frac{af_1}{a - f_1}$$

This distance is positive if the object is imaged behind the first lens. The image will be viewed through the second lens, with the pattern distance now equal to $-a'$. The second lens displays the object as

$$a'' = \frac{-a'f_2}{-a' - f_2} = \frac{-\left(\frac{af_1}{a - f_1}\right)f_2}{-\frac{af_1}{a - f_1} - f_2} = \frac{af_1f_2}{af_1 + af_2 - f_1f_2} = \frac{aF}{a - F}$$

where $\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$. So we derived an interesting relation that for two lenses in close vicinity to each other, their total optical power is the sum of the optical powers of the two lenses.

The relationship can be iterated and it is evident that the total optical power of all Jarda's lenses in a row is

$$\frac{1}{F} = \frac{1}{f} \left(\frac{1}{1} + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \right) = \frac{2}{f}$$

From here $F = \frac{f}{2}$. The focal length of this lens series is thus half that of the first lens. We can calculate the infinite sum above for example from the relation for an infinite geometric series. The set of these lenses will therefore image an object located at a distance f in front of it to a distance

$$x = \frac{f \frac{f}{2}}{f - \frac{f}{2}} = f$$

Now let's move on to the image on the concave mirror. It has a focal length equal to one half of the radius of curvature. For a point with distance $-f$ (the negative sign is due to the fact that the lens system imaged the object behind the mirror) the image will be in distance

$$x' = \frac{-x \frac{f}{2}}{-x - \frac{f}{2}} = \frac{f}{3}$$

Now the object is shown at a distance $x' = \frac{f}{3}$ in front of the mirror.

Further imaging is again performed using a system of lenses. The object is now located $x' = -\frac{f}{3}$ behind the lens system. For the last time we use the imaging equation

$$x'' = \frac{x' \frac{f}{2}}{x' - \frac{f}{2}} = \frac{\frac{-f}{3} \frac{f}{2}}{-\frac{1}{3} - \frac{1}{2}} = \frac{f}{5}$$

2. Ans (2)

$$\lambda_1 = 3000 \text{ \AA}$$

$$\lambda_2 = 6000 \text{ \AA}$$

$$E_1 = \frac{hc}{\lambda_1} = 4.14 \text{ eV},$$

$$E_2 = \frac{hc}{\lambda_2} = 2.07 \text{ eV}$$

$$(4.14) \text{ eV} - W = \frac{1}{2}mv_1^2 \quad \dots(i)$$

$$(2.07) \text{ eV} - W = \frac{1}{2}mv_2^2 \quad \dots(ii)$$

$$\frac{v_1}{v_2} = 3 \quad \dots(iii)$$

Equations (i), (ii) and (iii), we get

$$W = 1.81 \text{ eV}$$

6. Ans (1)

$$\text{Gives } \frac{dq'}{dq} = -\left(\frac{q'}{q}\right)^2$$

Thus for a small deviation of the object

distance (depth of field), Δq , the deviation of

the image distance, is

$$|\Delta q'| \approx |\Delta q| \left(\frac{q'}{q}\right)^2$$

By geometry,

$$\frac{\Delta q'}{\ell} = \frac{(q' + \Delta q')}{D} \approx \frac{q'}{D}$$

where D is the diameter of the lens. As $q \gg f$,

$q' \approx f$, we obtain

$$\Delta q \approx \frac{\ell q'}{D} \left(\frac{q}{q'}\right)^2 \approx \frac{\ell f}{D} \left(\frac{q}{f}\right)^2$$

PART-2 : CHEMISTRY

SECTION-I (i)

4. Ans (B)

Sucrose D(+) glucose + D(-) Fructose

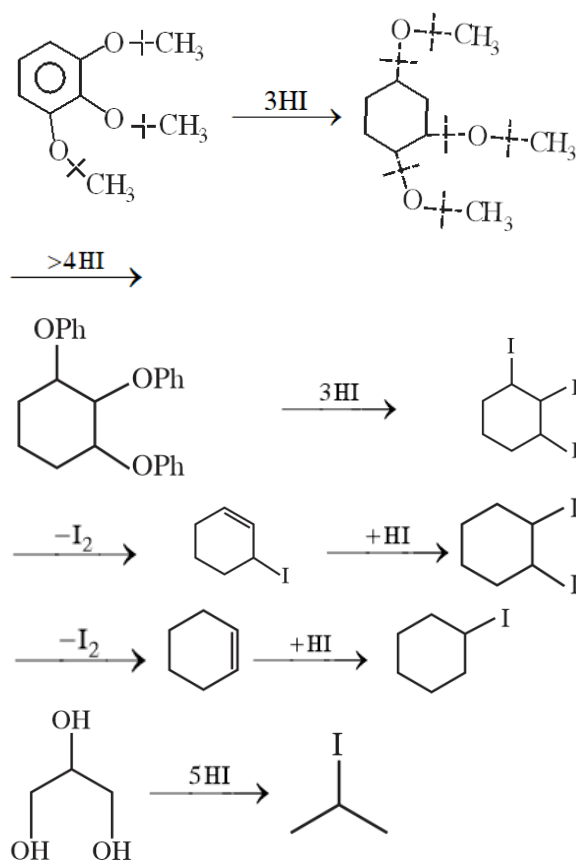
$$[\alpha] = +66.5^\circ \quad [\alpha] = +52.5^\circ \quad [\alpha] = -92.8^\circ$$

Thus rotation changes from positive to negative after hydrolysis. Due to this reason hydrolysis of sucrose is known as inversion and mixture after hydrolysis is known as invert sugar.

PART-2 : CHEMISTRY

SECTION-I (ii)

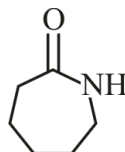
5. Ans (B,C,D)



PART-2 : CHEMISTRY

SECTION-II

2. Ans (2)



3. Ans (4)

Asp, Gly, Ala, Val

4. Ans (2)

Rate of dehydration reaction of alcohol depends upon stability of carbocation formed in rate determining step.

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (B)

Clearly α is repeated root of $f(x) = 0$

$$\lim_{x \rightarrow \alpha} \frac{f(x)}{f'(x)} \left(\frac{f'(x)}{f(x)} - \left\{ \frac{f'(x)}{f(x)} \right\} \right)$$

$$\Rightarrow \lim_{x \rightarrow \alpha} \left(1 - \frac{f'(x)}{f'(x)} \left\{ \frac{f'(x)}{f(x)} \right\} \right) = 1 - 0 = 1$$

$$\left(\lim_{x \rightarrow \alpha} \frac{f(x)}{f'(x)} = 0 \text{ and } \left\{ \frac{f'(x)}{f(x)} \right\} \text{ is bounded} \right.$$

function)

2. Ans (A)

$$\sin^{-1} \left\{ \frac{1}{2} \left(\frac{1}{x} + x \right) \right\} \text{ check } f(1^-) = \sin^{-1} 1^+,$$

$$f(1) = \sin^{-1} 1, f(1^+) = \sin^{-1} 1^-$$

Hence function is discontinuous so non differentiable

3. Ans (C)

$$f(x) = \frac{2(x^4 - 7x^2 - 4x + 23) + 3}{x^4 - 7x^2 - 4x + 23}$$

$$= 2 + \frac{3}{x^4 - 7x^2 - 4x + 23}$$

$$= 2 + \frac{3}{x^4 - 8x^2 + 16 + x^2 - 4x + 4 + 3}$$

$$= 2 + \frac{3}{(x^2 - 4)^2 + (x - 2)^2 + 3}$$

Range of $f(x) = (2, 3]$

4. Ans (A)

$$f(\theta) = \frac{(2 \cos \theta - 1)(2 \cos 2\theta - 1)(2 \cos 4\theta - 1) \dots (2 \cos 2^{n-1}\theta - 1)}{2 \cos 2^n \theta + 1}$$

for $n \in \mathbb{N}$

$$= \frac{(2 \cos \theta + 1)(2 \cos \theta - 1)(2 \cos 2\theta - 1)(2 \cos 4\theta - 1) \dots (2 \cos 2^{n-1}\theta - 1)}{(2 \cos \theta + 1)(2 \cos 2^n \theta + 1)}$$

$$\cos \theta \neq -\frac{1}{2} \text{ as } \theta \neq 2m\pi \pm \frac{2\pi}{3} \text{ for } m \in \mathbb{I}$$

$$= \frac{(4 \cos^2 \theta - 1)(2 \cos 2\theta - 1) \dots (2 \cos 2^{n-1}\theta - 1)}{(2 \cos \theta + 1)(2 \cos 2^n \theta + 1)}$$

$$= \frac{(2 \cos 2\theta + 1)(2 \cos 2\theta - 1) \dots (2 \cos 2^{n-1}\theta - 1)}{(2 \cos \theta + 1)(2 \cos 2^n \theta + 1)}$$

$$= \frac{1}{2 \cos \theta + 1}$$

$$\Rightarrow f\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2} + 1} = \sqrt{2} - 1$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (A,C,D)

$$\ell n(L) = \lim_{n \rightarrow \infty} \frac{1}{n} \ell n \left(\frac{f\left(a + \frac{1}{n}\right)}{f(a)} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n^2} \left[\frac{\ell n f\left(a + \frac{1}{n}\right) - \ell n f(a)}{\frac{1}{n}} \right]$$

$$= 0 [\ell n(f(x)) \text{ is differentiable}]$$

(C) Take log and apply L-hospital Rule.

6. Ans (A,B,D)

For (A) $(x-1)^2 + 4 = -\pi^x$ (not possible) $\cot^{-1}x$
 $= 3.25$ (not possible)

For (B) $\cot^{-1}x + \operatorname{sgn}e^x = 1 \Rightarrow \cot^{-1}x + 1 = 1$

$\Rightarrow \cot^{-1}x = 0 \Rightarrow$ not possible

$$x^4 - 2x^2 \sin^2 \frac{\pi}{2} x + 1 = 0 \text{ i.e. } x^4 + 1 = 2x^2 \sin^2 \frac{\pi x}{2}$$

$$\Rightarrow x^4 + 1 = 2x^2 \left(1 - \cos^2 \frac{\pi x}{2}\right)$$

$$\Rightarrow x^4 + 1 = 2x^2 - 2x^2 \cos^2 \frac{\pi x}{2}$$

$$\Rightarrow x^4 + 1 - 2x^2 = -2x^2 \cos^2 \frac{\pi x}{2}$$

$$\Rightarrow (x^2 - 1)^2 = -2x^2 \cos^2 \frac{\pi x}{2}$$

$$\text{i.e. } (x^2 - 1)^2 + 2x^2 \cos^2 \frac{\pi x}{2} = 0$$

sum of two perfect squares is 0 if each is zero

i.e. $x = \pm 1$ i.e. two solutions.

$$\tan\left(x + \frac{\pi}{6}\right) = 2 \tan x \Rightarrow \frac{\tan x + \tan \frac{\pi}{6}}{1 - \tan x \tan \frac{\pi}{6}} = 2 \tan x$$

Let $\tan x = y$

$$\frac{y + \frac{1}{\sqrt{3}}}{1 - \frac{y}{\sqrt{3}}} = 2y \Rightarrow \frac{\sqrt{3}y + 1}{\sqrt{3} - y} = 2y$$

$$\text{i.e. } \sqrt{3}y + 1 = 2\sqrt{3}y - y^2$$

$$\text{i.e. } y^2 - \sqrt{3}y + 1 = 0 \text{ its } D < 0 \text{ so No solution.}$$

7. Ans (B,C,D)

$$f(x) + 2f(1-x) = x^2 + 1 \quad \forall x \in \mathbb{R}$$

$$x \rightarrow 1-x$$

$$f(1-x) + 2f(x) = (1-x)^2 + 1 \quad \forall x \in \mathbb{R}$$

$$\Rightarrow 4f(x) + 2f(1-x) = 2(1-x)^2 + 2 \quad \dots(1)$$

$$f(x) + 2f(1-x) = x^2 + 1 \quad \dots(2)$$

Subtracting Eq. (2) from Eq. (1)

$$\Rightarrow 3f(x) = 2[x^2 - 2x + 1] - x^2 + 1 = x^2 - 4x + 3$$

$$f(x) = \frac{1}{3}(x^2 - 4x + 3)$$

$$f(x) = \frac{1}{3}((x-2)^2 - 1)$$

$$\text{Range of } f(x) = \left[-\frac{1}{3}, \infty\right)$$

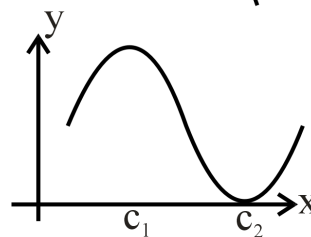
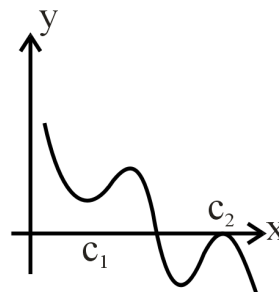
PART-3 : MATHEMATICS

SECTION-I (iii)

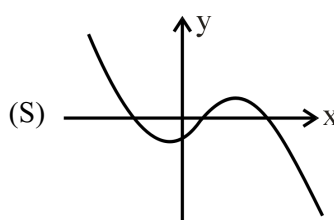
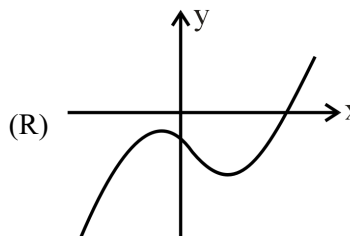
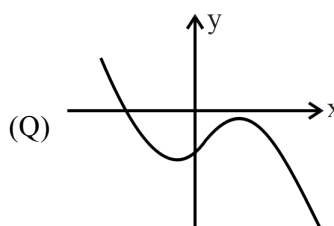
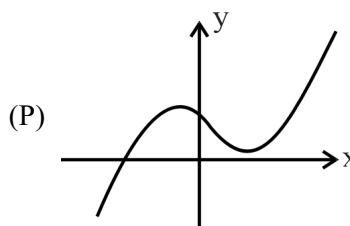
8. Ans (C)

for $f'(c_1) - f'(c_2) > 0$

$f'(c_1) - f'(c_2) < 0$



9. Ans (C)



10. Ans (B)

$$(P) f_1(x) = x^2 \operatorname{sgn}(x) = \begin{cases} x^2 & x \geq 0 \\ -x^2 & x < 0 \end{cases}$$

$$f_1'(x) = \begin{cases} 2x & x > 0 \\ 2x & x < 0 \end{cases}, f_1'(0) = 0$$

$$f_1''(x) = \begin{cases} -2 & x < 0 \\ 2 & x > 0 \end{cases}$$

So first derivative exists at $x = 0$ but second derivative does not exist there

$$(Q) f_2(x) = \int_0^x t^2 \sin\left(\frac{1}{t}\right) dt = f_2(0) = 0$$

$$f_2'(x) = x^2 \sin \frac{1}{x}, x \neq 0, g'(0) = 0$$

$$f_2''(0) = 0 \text{ but } \lim_{x \rightarrow 0} f_2''(x) \text{ does not exist}$$

$$(R) f_3(x) = x^{-\frac{1}{3}} |\sin x|, f_3(0) = 0, \lim_{x \rightarrow 0} f_3(x) = 0$$

$$f_3'(0^+) = \lim_{x \rightarrow 0} \frac{x^{-\frac{1}{3}} \sin x}{x} \text{ does not exist}$$

$$(S) f_4(x) = \begin{cases} 0 & -1 < x \leq 0 \\ -x^3 & 0 < x < 1 \end{cases}$$

Second derivative exists and continuous at $x = 0$

11. Ans (A)

$$\sin^{-1}\left(\frac{4x}{4+x^2}\right) = \begin{cases} -\pi - 2\tan^{-1}\left(\frac{x}{2}\right) & ; x < -2 \\ 2\tan^{-1}\left(\frac{x}{2}\right) & ; -2 \leq x \leq 2 \\ \pi - 2\tan^{-1}\left(\frac{x}{2}\right) & ; x > 2 \end{cases}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (1)

$$x^3 + 3x^2 + 3x + 1 - (x-6)^{1/3} + 7 = 0$$

$$\Rightarrow (x+1)^3 + 6 + 1 - (x-6)^{1/3} = 0$$

$$\Rightarrow (x+1)^3 + 6 = (x-6)^{1/3} - 1$$

$$\Rightarrow \text{Solution of } f(x) = f^{-1}(x)$$

$$\Rightarrow f(x) = x \Rightarrow x^3 + 3x^2 + 2x + 7 = 0$$

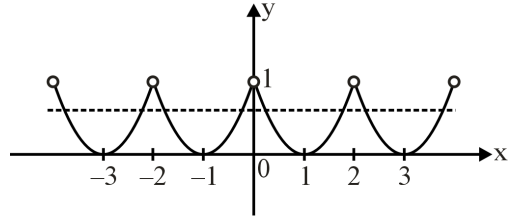
$\Rightarrow$ Only one solution

2. Ans (8)

$$\left(x - 2 \cdot \left[\frac{x}{2}\right] - 1\right)^2 = \frac{1}{2} \Rightarrow \left(x - 2 \left(\frac{x}{2} - \left\{\frac{x}{2}\right\}\right) - 1\right)^2 = \frac{1}{2}$$

$$\Rightarrow \left(2 \left\{\frac{x}{2}\right\} - 1\right)^2 = \frac{1}{2}$$

Let $f(x) = 2 \cdot \left\{\frac{x}{2}\right\} - 1$ which is periodic with period 2.



Now,

$$f(x) = 2 \left\{\frac{x}{2}\right\} - 1 = \begin{cases} (x-1)^2 & 0 \leq x < 2 \end{cases}$$

$\Rightarrow$ Graph of $f(x)$ is adjacent.

Now solution of $f(x) = \frac{1}{2}$ intersects the curve at

8 points in $x \in [-4, 4]$

3. Ans (1)

$$f : A \rightarrow \left[-\frac{1}{2}, 0\right]$$

$$f^3(x) + x^3 + 1 = 3x \cdot f(x)$$

$$\Rightarrow f(x) = -x - 1 \left(A \rightarrow \left[-1, -\frac{1}{2}\right] \right)$$

$$\therefore a^3 + b^3 + c^3 = 3abc$$

$$\Rightarrow a + b + c = 0 \quad (a \neq b \neq c)$$

$$f(f(x)) = x \text{ \& } f^{-1}(x) = -x - 1$$

$$x^2 = \frac{3}{2}(-x - 1) + 1$$

$$2x^2 + 3x + 1 = 0$$

$$= -\frac{1}{2} \text{ \& } -1 \text{ (reject)}$$

because $A \in \left[-1, -\frac{1}{2}\right]$ and domain of $f^{-1}(x)$ is

4. **Ans (6)**

$$\lim_{x \rightarrow 0} \left(3 - \frac{P(x)}{x} \right) = 27$$

$\Rightarrow P(x)$ has no constant term

$$\text{let } P(x) = ax^4 + bx^3 + cx^2 + dx$$

$$\Rightarrow 3 - d = 27 \Rightarrow d = -24$$

$$P(x) = ax^4 + bx^3 + cx^2 - 24x$$

$$P'(2) = 0, p(1) = -9, p'''(2) = 0$$

$$P'(x) = 4ax^3 + 3bx^2 + 2cx - 24$$

$$P''(x) = 12ax^2 + 6bx + 2c$$

$$P'''(x) = 24ax + 6b$$

$$a + b + c - 24 = -9$$

$$\Rightarrow a + b + c = 15 \quad \dots(1)$$

$$P'(2) = 0$$

$$\Rightarrow 4a(8) + 3b(4) + 2c(2) - 24 = 0$$

$$\Rightarrow 8a + 3b + c = 6 \quad \dots(2)$$

$$P'''(2) = 0$$

$$\Rightarrow 24a(2) + 6(b) = 0$$

$$\Rightarrow 8a + b = 0 \quad \dots(3)$$

Solving (1), (2) & (3)

$$a = 1, b = -8, c = 22$$

$$\Rightarrow P(x) = x^4 - 8x^3 + 22x^2 - 24x$$

$$P'(x) = 4x^3 - 24x^2 + 44x - 24$$

$$= 4[x^3 - 6x^2 + 11x - 6]$$

$$P'(x) = 4[(x-1)(x-2)(x-3)]$$

$$P''(x) = 4[3x^2 - 12x + 11] > 0 \quad \forall x \in [3, 4]$$

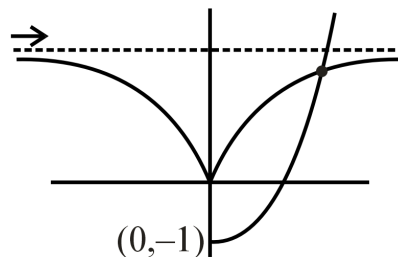
$$\Rightarrow P'(x) = 4[(4-1)(4-2)(4-3)]$$

$$= 4[(3)(2)(1)]$$

$$= 24 = 4M$$

$$\Rightarrow M = 6$$

5. **Ans (2)**



6. **Ans (324)**

Number of ways of discontinuity at a point $x = c$

are 3.

(A) LHL at $x = c$ is equal to $f(c)$ but not equal to

RHL at $x = c$

(B) LHL at $x = c$ is not equal to $f(c)$ but equal to

RHL at $x = c$

(C) LHL at $x = c$ is neither equal to $f(c)$ nor

equal to RHL at $x = c$ (where $f(c)$) is equal to

RHL at $x = c$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

PRE - SCORE

20-11-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS, TNAS, TAAS & TTAS)

ANSWER KEY

PAPER-2 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	A	C	C	B		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,C	A,B	A,B			
SECTION-II (i)	Q.	1	2	3	4		
	A.	1.00	10.00	20.00	4.00		
SECTION-II (ii)	Q.	5	6	7	8	9	10
	A.	27	10	22	5	25	1

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	A	A	A		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,B,C,D	A,B,D	A			
SECTION-II (i)	Q.	1	2	3	4		
	A.	8.00	15.00	8.00	9.00		
SECTION-II (ii)	Q.	5	6	7	8	9	10
	A.	12	5	5	4	6	6

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	B	C	C		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,D	A,B,D	A,C			
SECTION-II (i)	Q.	1	2	3	4		
	A.	2.00	2.00	2.00	1.00		
SECTION-II (ii)	Q.	5	6	7	8	9	10
	A.	4	5	2	20	8	6

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

2. Ans (C)

Radioactive decay obeys the equation

$$N_x = N_0 e^{-\lambda_x t}$$

where N_0 is the initial number of particles and

$\lambda_r = \frac{\ln 2}{T}$ is the decay constant. The number of

particles that decay in a time dt is then

$$-dN_r = -N_r dt = \lambda_r N_0 e^{-\lambda_r t} dt = \lambda_r N dt$$

However, in our case, the radioactive solution is

also being diluted by the clean water coming to

the well. After the time dt , the volume $q \backslash dt$ will

flow through the well. The number of

radioactive particles leaving the well that way is

$$\frac{qvdt}{V} N = -dN_v$$

Therefore, the equation describing the change in

the number of radioactive particles in the well is

$$dN = dN_r + dN_v = - \left(\lambda_r + \frac{qv}{V} \right) N dt = -\lambda N dt$$

We now see that the number of particles in the

well will again be an exponential function, but

with a different constant. The activity can be

calculated as $A = \lambda_r N$, or $A = \lambda_r N_0 e^{-\lambda t}$

From this, we get the resulting time

$$t = -\frac{1}{\lambda} \ln \left(\frac{A}{\lambda_r N_0} \right) = \frac{1}{\frac{\ln 2}{T} + \frac{qv}{V}} \ln \left(\frac{N_0}{A} \frac{\ln 2}{T} \right)$$

3. Ans (C)

Once a gamma ray of energy E is produced and

begins propagating through the material, it is also

attenuated, now with a coefficient of μ_2 . Let the

intensity of this radiation dJ_0 occur in the dx -

neighbourhood of the x coordinate. According to

the formula above, this intensity drops after

exiting the material to

$$dJ = dJ_0 e^{-\mu_2(t-x)}$$

since the radiation has now passed a distance $t - x$

in the material.

But the resulting intensity dJ_0 is proportional to

$I(x)$ and the length dx through some

proportionality coefficient a , which we write as

$$dJ_0 = aI(x)dx = aI_0 e^{-\mu_1 x} dx$$

Substituting into the previous equation and

integrating over the entire thickness of the

material gives the total radiation intensity with

energy E as

$$J = aI_0 \int_0^t e^{-\mu_1 x} e^{-\mu_2 x(t-x)} dx = \int_0^t e^{(\mu_2 - \mu_1)x} dx = \frac{aI_0 e^{-\mu_2 t}}{\mu_2 - \mu_1} (e^{(\mu_2 - \mu_1)t} - 1)$$

The maximum intensity is found by a derivative

with respect to t

$$\frac{dJ}{dt} = \frac{aI_0}{\mu_2 - \mu_1} (-\mu_2 e^{-\mu_2 t} (e^{(\mu_2 - \mu_1)t} - 1) + e^{-\mu_2 t} (\mu_2 - \mu_1) (e^{(\mu_2 - \mu_1)t}))$$

which we set equal to zero. We get

$$0 = -\mu_2 e^{(\mu_2 - \mu_1)t_m} + \mu_2 + (\mu_2 - \mu_1) e^{(\mu_2 - \mu_1)t_m},$$

from which

$$\mu_1 e^{(\mu_2 - \mu_1)t_m} = \mu_2,$$

and now we can simply express the thickness for

which the intensity is external as

$$t_m = \frac{1}{\mu_2 - \mu_1} \ln \left(\frac{\mu_2}{\mu_1} \right)$$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (A,C)

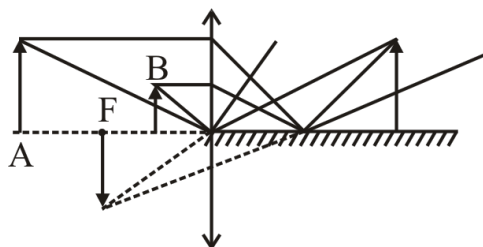
For object A,

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f} \Rightarrow \frac{1}{2f} + \frac{1}{v} = \frac{1}{f} \quad (1 \text{ point})$$

$$\Rightarrow v = 2f \quad (1 \text{ point})$$

$$\text{and } m = \left| \frac{v}{u} \right| = 1 \quad (1 \text{ point})$$

$$\Rightarrow \text{real, up} \quad (1 \text{ point})$$



For object B,

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f} \Rightarrow \frac{1}{f/2} + \frac{1}{v} = \frac{1}{f} \quad (1 \text{ point})$$

$$\Rightarrow v = -f \quad (1 \text{ point})$$

$$m = \left| \frac{v}{u} \right| = 2 \quad (1 \text{ point})$$

$$\Rightarrow \text{virtual, inverted} \quad (1 \text{ point})$$

6. Ans (A,B)

A point on the sphere's equator moves at a speed ωR , where ω is the angular velocity of rotation.

Setting $\omega R = c$ gives $\omega = c/R$.

The spinning sphere can be thought of as a stack of infinitesimal current loops, all of which have a magnetic moment pointing in the same direction.

Consider making a thin, circular slice of the sphere's surface, corresponding to polar angles in the range $(\theta, \theta + d\theta)$. This slice has a radius r

$\sin\theta$, so that the surface area of the slice is

$$ds = 2\pi R \sin\theta R d\theta$$

The charge of the slice is

$$dQ = -\frac{q ds}{4\pi R^2} = -\frac{q \sin\theta}{2}$$

Since the charge dQ moves around the rotation

axis one time per period $t = \frac{2\pi}{\omega}$, the

corresponding current is

$$dI = \frac{dQ}{T} = -\frac{\omega q \sin\theta}{4\pi}$$

The magnitude of the magnetic moment of this

slice is

$$d\mu = \pi (R \sin\theta)^2 |dI| = \frac{1}{4} q \omega R^2 \sin^3\theta d\theta$$

Using the provided integral, the total magnetic moment is

$$\mu = \int_0^\pi \frac{1}{4} q \omega R^2 \sin^3\theta d\theta = \frac{1}{3} q c R$$

PART-1 : PHYSICS

SECTION-II (i)

1. Ans (1.00)

$$\frac{1}{f_{eq}} = \left[\frac{3}{2} - 1 \right] \left[\frac{1}{R} - \frac{1}{-R} \right]; f_{eq} = R:$$

∴ Object should be placed at $2f_{eq} = R$

2. Ans (10.00)

$$\frac{1}{f} = \frac{1}{f_{glass}} - \frac{1}{f_{water}} = \left(\frac{3}{2} - 1 \right) \left(\frac{1}{R} - \frac{1}{-R} \right) + \left(\frac{4}{3} - 1 \right) \left(\frac{1}{-R} - \frac{1}{\infty} \right) = \frac{1}{R} - \frac{1}{3R}$$

$f = \frac{3}{2}R$ image will coincide with object if it placed at focus because if object is placed at this position ray will be incident normally on mirror.

$$\frac{3R}{2} = 15 = 10 \text{ cm}$$

PART-1 : PHYSICS

SECTION-II (ii)

5. Ans (27)

Let us denote the object distance from the connecting lens as a . According to the Gaussian lens formula, the object will appear through the lens at a distance

$$a' = \frac{af}{a-f}$$

The distance of this image1 from the second lens is $a_2 = d - a'$. We will use the Gaussian lens formula for the second time and get

$$a'_2 = \frac{-a_2 f}{a_2 + f} = \frac{-(d - a')f}{d - \frac{af}{a-f} + f}$$

According to the problem statement, the position of the image is equal to the position of the object, therefore $a'_2 = -d - a$. So we are trying to solve the equation

$$= \frac{-(da - df - af)f}{da - df - f^2} = -d - a$$

from which we get

$$a^2 + (d - 2f)a - df = 0$$

The solution to this equation is

$$a = \frac{-d + 2f \pm \sqrt{d^2 + 4f^2}}{2}$$

6. Ans (10)

Notice that some of the rays impinge on the hypotenuse at a greater angle than the critical angle (e.g. a vertical ray). Therefore, the hypotenuse of the prism acts as a mirror and we can consider the mirror image of the object which is entirely located at 12.5cm left of the mid-point of the hypotenuse.

Now, we have neglected the effect of refraction at the other sides of the prism. To account for this, simply map the mirror images of those edges as well and bend the rays of the mirror image of the object when they cross these virtual edges. The object is then effectively located 8cm on the left of a slab with refractive index 1.5 and thickness 9cm. Since passing through a slab forms another image which is tantamount to reducing the object distance by $t \left(1 - \frac{1}{n} \right)$, where t is the thickness of the slab, the effective object distance to the first lens is

$$8 + 9 - 9 \left(1 - \frac{1}{\frac{3}{2}} \right) + 6 = 20 \text{ cm}$$

which coincides with the front focal point of the first lens. Therefore, the image of the first lens is at infinity. The final image is then formed at

$$v = -10 \text{ cm}$$

8. Ans (5)

In this problem we can use the conservation of energy: the binding energy E_s of chlorine's

molecule is equal to the difference where the

mechanical energy $E_1 = m_{Cl} \frac{v_1^2}{2} + m_{Cl} \frac{v_2^2}{2}$ of

chlorine molecules after division is subtracted

from photon energy $\frac{hc}{\lambda}$ and initial mechanical

energy $E_0 = 2m_{Cl} \frac{v^2}{2}$.

Let the x-axis be parallel to the initial

movement of the molecule. We know that one

of the atoms moved perpendicularly to the x-

axis after the division. This means that $v_{1x} = 0$

and $v_{1y} = u$. From the conservation of

momentum we get the components of the other

atom's velocity $v_{2x} = 2v$ and $v_{2y} = -u$. Now we

can write down the conservation of energy:

$$\frac{hc}{\lambda} = m_{Cl} v^2 = E_s + m_{Cl} \frac{u^2}{2} + m_{Cl} \frac{4u^2 + u^2}{2}$$

$$= E_s + 2m_{Cl} v^2 + m_{Cl} u^2$$

From this we can express E_s considering that

$$m_{Cl} = \frac{\mu_{Cl}}{N_A} \text{ where } \mu_{Cl} = 35.10^{-3} \text{ kg/mol.}$$

$$E_s + \frac{hc}{\lambda} - m_{Cl} v^2 - m_{Cl} u^2 = \frac{hc}{\lambda} - \frac{\mu_{Cl}}{N_A} (v^2 + u^2)$$

$$= 4.0 \times 10^{-19} \text{ J} = 2.5 \text{ eV}$$

9. Ans (25)

Let us treat the interaction between the proton

and the electron as an elastic collision. Let the

proton's initial velocity be u , final velocity v and

the electron's velocity momentarily before and

after the collision w . Initially we can ignore that

the electron belongs in the hydrogen's atom.

From the conservation of energy

$$\frac{1}{2} m_p u^2 = \frac{1}{2} m_p v^2 + \frac{1}{2} m_e w^2 \text{ and conservation of}$$

$$\text{momentum } m_p \vec{u} = m_p \vec{v} + m_e \vec{w}. \text{ From the}$$

scalar multiplication of the latter equation with

$$\text{itself we get } m_p^2 u^2 = m_p^2 v^2 + m_e^2 w^2 + 2m_p m_e \vec{v} \cdot \vec{w}$$

. We replace u from the conservation of energy.

$$m_p (m_p v^2 + m_e w^2) = m_p^2 v^2 + m_e^2 w^2 +$$

$$2m_p m_e \vec{v} \cdot \vec{w}, \text{ therefore } (m_p - m_e) w^2 = 2m_p \vec{v} \cdot \vec{w}$$

$$= 2m_p v w \cos \theta \text{ where } \theta \text{ is the angle between the}$$

vectors $\vec{v}$ and $\vec{w}$. The removal of the electron

from the hydrogen's atom is possible if the

electron's total energy $E > 0$. Thus, we try to

maximize the electron's kinetic energy and also

the velocity w after the collision.

$$W = \max \left(\frac{2m_p v}{m_p - m_e} \cos \theta \right) = \frac{2m_p v}{m_p - m_e}.$$

From the energy's equation we get

$$K_0 = \frac{1}{2} m_p u^2 = \frac{1}{2} m_p \left(\frac{(m_p - m_e) W}{2m_p} \right)^2$$

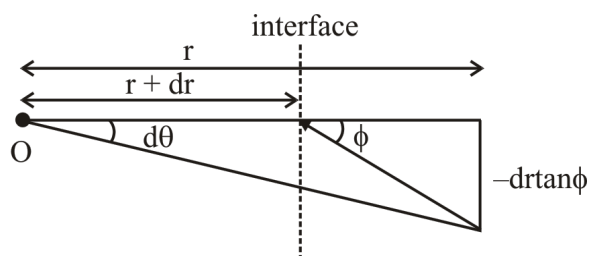
$$+ \frac{1}{2} m_e W^2 = \frac{(m_p + m_e)^2}{8m_p} W^2.$$

$$\text{To ionize } K_e = \frac{1}{2} m_e W^2. E_0 \text{ meaning}$$

$$K_0 > \frac{E_0 (m_p + m_e)^2}{4m_p m_e} = 6.25 \text{ keV.}$$

10. Ans (1)

In figure, let the origin O be located at the center of the circle and define the coordinates of the light ray as (r, θ) . Since the refractive index is strictly a function of r , the interfaces are concentric circles. Let ϕ be the angle of incidence of the light ray as it propagates through this circle.



Consider the propagation of the light ray from radial coordinate r to $r + dr$ where $dr < 0$.

Evidently, the angular change $d\theta$ is given by

$$r d\theta = -dr \tan \phi$$

By Snell's law, $n \sin \phi$ is constant.

$$n \sin \phi = n(R) \sin \frac{\pi}{4} = \sqrt{2} \cdot \frac{1}{\sqrt{2}} = 1$$

$$\sin \phi = \frac{1}{n}$$

$$\tan \phi = \frac{\frac{1}{n}}{\sqrt{1 - \frac{1}{n^2}}} = \frac{1}{\sqrt{n^2 - 1}}$$

for $0 \leq \phi < \frac{\pi}{2}$. Substituting the expression for $n(r)$,

$$\tan \phi = \frac{r}{R}$$

Substituting this into the first equation.

$$\int_{\theta_0}^{\theta_0 + \Delta\theta} d\theta = \int_R^0 \frac{\tan \phi}{r} dr = \int_R^0 \frac{1}{R} dr$$

$$|\Delta\theta| = 1 \text{ radian}$$

PART-2 : CHEMISTRY

SECTION-II (i)

2. Ans (15.00)

$$x = 4, y = 6, z = 5$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (B)

$$x^2 - x - 2 \neq 0 \text{ and } \left[x + \frac{1}{2} \right] > 0, \left[x + \frac{1}{2} \right] \neq 1$$

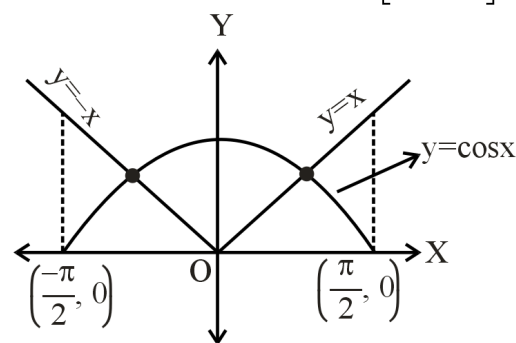
$$\Rightarrow \left[x + \frac{1}{2} \right] \geq 2$$

$$x^2 - x - 2 \neq 0 \Rightarrow x \neq -1, 2 \text{ and } \left[x + \frac{1}{2} \right] \geq 2$$

$$\Rightarrow x \geq \frac{3}{2} \Rightarrow x \in \left[\frac{3}{2}, 2 \right) \cup (2, \infty)$$

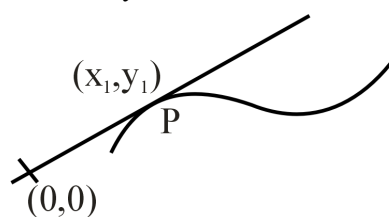
2. Ans (B)

$$\text{As, } |\sin^{-1}(\sin x)| = |x|, \text{ for } x \in \left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$$



3. Ans (C)

$$\text{We have } y = x^3 + x + 16$$



$$\left. \frac{dy}{dx} \right|_P = 3x^2 + 1 = 3x_1^2 + 1$$

$$\therefore 3x_1^2 + 1 = \frac{y_1}{x_1}$$

$$\Rightarrow 3x_1^2 + 1 = \frac{x_1^3 + x_1 + 16}{x_1}$$

$$\Rightarrow 3x_1^3 + x_1 = x_1^3 + x_1 + 16$$

$$\Rightarrow 2x_1^3 = 16 \Rightarrow x_1 = 2$$

$$\text{Hence slope of line } \ell = 3(2)^2 + 1 = 13$$

4. Ans (C)

$$\lim_{x \rightarrow 0} \frac{x^3 \cdot \frac{\tan^3 x}{x^3} + 8x^3 \cdot \frac{\tan^3 2x}{(2x)^3} + 27x^3 \cdot \frac{\tan^3 3x}{(3x)^3}}{64x^3 \cdot \frac{\tan^3 4x}{(4x)^3} + 125x^3 \cdot \frac{\tan^3 5x}{(5x)^3} + 216x^3 \cdot \frac{\tan^3 6x}{(6x)^3}}$$

$$= \frac{1 + 8 + 27}{64 + 125 + 216} = \frac{36}{405} = \frac{4}{45}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (A,D)

$$\lim_{x \rightarrow 0} x \sin\left(\frac{\pi}{x}\right) = 0$$

(A) So for $k = 1$, $f(x)$ has removable discontinuity at $x = 0$

(B) For $k = 0$, $f(x)$ is continuous at $x = 0$

6. Ans (A,B,D)

$$\frac{dy}{dx} = \frac{6x^2}{2} - 4x + c$$

$$f'(x) = 3x^2 - 4x + c$$

$$0 = -1 + c \Rightarrow c = 1 \quad (\because f'(1) = 0)$$

$$f(x) = x^3 - 2x^2 + x + k$$

$$5 = k \quad (\because f(1) = 5)$$

$$f(x) = x^3 - 2x^2 + x + 5$$

$$f'(x) = 3x^2 - 4x + 1$$

$$= (3x - 1)(x - 1)$$

7. Ans (A,C)

$$\cos^{-1}(\cot(\tan^{-1}x)) = 0, \pi$$

$$\Rightarrow \cot(\tan^{-1}x) = 1, -1$$

$$\Rightarrow \tan^{-1}x = -\frac{\pi}{4}, \frac{\pi}{4}$$

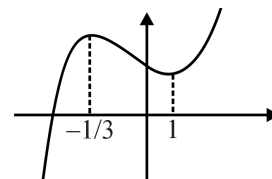
$$\Rightarrow x = -1, 1$$

PART-3 : MATHEMATICS

SECTION-II (i)

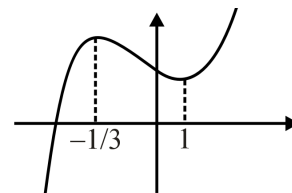
1. Ans (2.00)

$$f(x) = \begin{cases} x^3 - x^2 + x - 2 & x > 2 \\ x^3 - x^2 + 2 - x & 1 < x \leq 2 \\ x^2 - x^3 + 2 - x & 0 < x \leq 1 \\ x^3 - x^2 + 2 - x & x \leq 0 \end{cases}$$

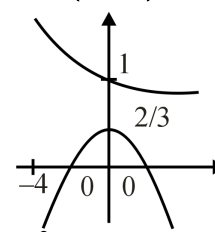


2. Ans (2.00)

$$f(x) = \begin{cases} x^3 - x^2 + x - 2 & x > 2 \\ x^3 - x^2 + 2 - x & 1 < x \leq 2 \\ x^2 - x^3 + 2 - x & 0 < x \leq 1 \\ x^3 - x^2 + 2 - x & x \leq 0 \end{cases}$$



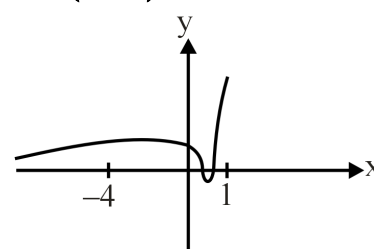
3. Ans (2.00)



$$(x^2 - 2)e^x = -3 \Rightarrow x^2 - 2 = -3e^{-x}$$

$$\Rightarrow \frac{2 - x^2}{3} = e^{-x}$$

4. Ans (1.00)



PART-3 : MATHEMATICS

SECTION-II (ii)

5. Ans (4)

$$y = (x^3 + 1) |x^2 - 1| + \sin|x + 1| \text{ is}$$

Not diff. at $x = 1, -1$

$$\text{so } \boxed{P = 2}$$

$$x^3 + x^2 - x + 3 = 0 \text{ have only one root}$$

$$\text{becomes } f\left(\frac{1}{3}\right) f(-1) > 0$$

$$\text{so } \boxed{Q = 1}$$

$$r = \sqrt{3} \tan\left(\frac{\pi}{3} - \frac{\pi}{6}\right)$$

$$r = \sqrt{3} \tan \frac{\pi}{6} = 1$$

6. Ans (5)

$f(x)$ should be CTS at $x = 0$ and $x = 2$

$$5 = \alpha \text{ and } 2\beta + \gamma = 4 + 10 + \alpha$$

$$\alpha = 5 \text{ and } 2\beta + \gamma = 19$$

$f(x)$ should be diff. at $x = 2$

$$2x + 5 \mid x = 2 = \beta$$

$$\boxed{\beta = 9}$$

$$\alpha = 5, \beta = 9, \gamma = 1, f'(1) = 2, f(3) = 9$$

7. Ans (2)

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{1 - \cos x \sqrt{\cos 2x}}{x^2} &= \lim_{x \rightarrow 0} \frac{1 - \cos^2 x \cos 2x}{(1 + \cos x \sqrt{\cos 2x}) x^2} \\ &= \lim_{x \rightarrow 0} \frac{1}{(1 + \cos x \sqrt{\cos 2x})} \cdot \lim_{x \rightarrow 0} \frac{1 - \cos^2 x \cos 2x}{x^2} \\ &= \frac{1}{2} \lim_{x \rightarrow 0} \frac{1 - \left(\frac{1 + \cos 2x}{2}\right) \cos 2x}{x^2} = \lim_{x \rightarrow 0} \frac{2 - \cos 2x - (\cos 2x)^2}{4x^2} \end{aligned}$$

$$= \lim_{x \rightarrow 0} (\cos 2x + 2) \cdot \lim_{x \rightarrow 0} \frac{1 - \cos 2x}{(2x)^2} = \frac{3}{2}$$

$$\Rightarrow \text{limit} = \frac{3}{2} + \frac{1}{2} = 2$$

8. Ans (20)

$$f(x) = \tan^{-1}(\cot x - 2 \cot 2x)$$

$$= \tan^{-1} \left(\frac{1}{\tan x} - \frac{1 - \tan^2 x}{\tan x} \right)$$

$$= \tan^{-1} \left(\frac{\tan^2 x}{\tan x} \right) = \tan^{-1}(\tan x)$$

$$\text{Now, } \sum_{r=1}^5 f(r) = f(1) + f(2) + f(3) + f(4) + f(5)$$

$$f(1) = 1$$

$$f(2) = 2 - \pi$$

$$f(3) = 3 - \pi$$

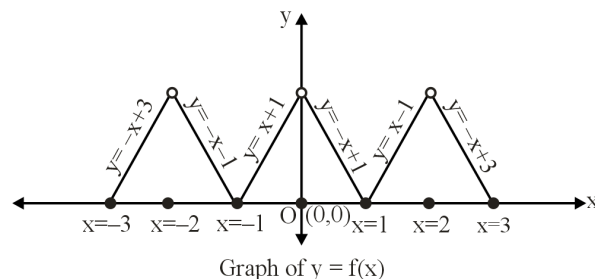
$$f(4) = 4 - \pi$$

$$f(5) = 5 - 2\pi$$

$$\Rightarrow \sum_{r=1}^5 f(r) = 15 - 5\pi = a - b\pi \Rightarrow a = 15, b = 5$$

So, value of $(a + b) = 20$

9. Ans (8)



10. Ans (6)

$$\left. \begin{aligned} x &= 6\cos^3 \theta \\ y &= 6\sin^3 \theta \end{aligned} \right\} \cdot \frac{dy}{dx} = \frac{-6.3\sin^2 \theta \cdot \cos \theta}{6.3\cos^2 \theta \cdot \sin \theta}$$

$$\frac{dy}{dx} = -\tan \theta$$

$$T : (y - 6\sin^3 \theta) = -\tan \theta (x - 6\cos^3 \theta)$$

$$N : (y - 6\sin^3 \theta) = \cot \theta (x - 6\cos^3 \theta)$$

$$T : x \sin \theta + y \cos \theta = 6 \sin \theta \cos \theta$$

$$N : x \cos \theta - y \sin \theta = 6 \cos 2\theta$$

$$p_1 = 6 \sin \theta \cos \theta = 3 \sin 2\theta$$

$$p_2 = 6 \cos 2\theta$$

$$\Rightarrow 4p_1^2 + p_2^2 = 36$$